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Neural Indicators of Error Processing and Intraindividual Variability in Reaction Time in 7 and 9 Year-Olds

ABSTRACT: Childhood is associated with improvements in task accuracy, response time, and reductions in intraindividual trial-to-trial variability in reaction times. The aims of this study were to investigate neural indicators of error monitoring to better understand the mechanisms underlying these cognitive developments in primary school aged children. Specifically, this study explored the development of error processing in 36 children aged 7 years and 41 children aged 9 years, as indexed by two electrophysiological indices of error processing, the error-related negativity (ERN) and the error positivity (Pe). Notably, the amplitude and latency of the ERN and Pe did not differ significantly between the age groups. However, intraindividual variability in response time (RT) was strongly related to ERN amplitude. These data suggest the utility of comparing neural and behavioral indicators of cognitive performance in children and uniquely highlight the importance of considering intraindividual variability in task performance in studies that explore error processing. © 2010 Wiley Periodicals, Inc. *Dev Psychobiol* 53: 256–265, 2011.

Keywords: error processing; intraindividual variability; event-related potentials; ERN; children; development

INTRODUCTION

The maturation of the frontal lobes of the brain is concomitant with children's cognitive development (Davidson, Amso, Anderson, & Diamond, 2006; Diamond, 2002; Giedd et al., 1999; Shaw et al., 2006; Sowell et al., 2004). Significant improvements in abilities considered dependent on the prefrontal cortex are found between the ages of 3 and 6 years, and 7 and 11 years

(Diamond, 2002). At around the age of 7, there are also marked improvements in children's executive functioning (Welsh, Pennington, & Grossier, 1991) and significant changes in information processing (Fry & Hale, 1996; Kail, 1991). Indeed, in our laboratory we find that improvements on a test of fluid intelligence (Cattell's Culture Fair of 'g'; Institute for Personality and Ability Testing, 1973) between the ages of 7 and 11 years are the same order of magnitude as those changes associated with normal ageing from 55 to over 80 years.

Of particular importance for the current study is evidence of age-related changes in accuracy, response speed, and trial-to-trial intraindividual variability during performance of elementary cognitive tasks (Davidson et al., 2006; Jensen, 1992; Kail, 1991; Li et al., 2004; Ridderinkhof, van der Molen, Band, & Bashore, 1997). Integral to these changes is the capacity to consistently detect and respond to errors through the ability to hold rules for correct task performance on-line, focus and switch attention, and coordinate responses to different types of stimuli in order to make moment-by-moment adjustments to behavior. We are interested in discovering

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some of the brain-based influences on these developmental changes. In one of the few brain-based studies to include younger children, a developmental trajectory of the error-related negativity (ERN), an electrophysiological correlate of error monitoring, was found in 7–18 year-olds (Davies, Segalowitz, & Gavin, 2004). There are two main goals of the current study: (1) to attempt to replicate the Davies et al. (2004) finding in a larger sample of 7- and 9-year-old children and (2) to test whether the ERN is related to error processing and response variability in this sample. In so doing we hope to further elucidate the nature of the relationship between the ERN and error processing in children.

Consistency of performance can be measured behaviorally by response time (RT) variability on cognitive tasks (Mathewson, Dywan, & Segalowitz, 2005; Unsworth, Redick, Lakey, & Young, 2010; Weissman, Roberts, Visscher, & Woldorff, 2006) as indexed typically by RT standard deviations (RTSD). Both RT and RTSD are reduced with increasing age during childhood (Davidson et al., 2006; Dougherty & Haith, 1997; Li et al., 2004). Nevertheless, individual differences in mean RT at each age continue to provide an index of general cognitive ability (Jensen, 1992). These behavioral measures not only offer descriptions of task efficiency, but also they are supported by theoretical accounts of how RT relates to cognitive ability (Anderson, 1992; Jensen, 1992, 2006). In this paper, we are exploring the potential for gathering brain-based evidence of, and explanations for, these behavioral observations of cognitive development in children.

Electrophysiological correlates of performance monitoring have indicated the existence of an error-monitoring mechanism in the brain. The discovery of an event-related potential (ERP) component associated with the occurrence of an error, the error negativity (Ne; Falkenstein, Hohnsbein, Hoorman, & Blanke, 1991), or the error-related negativity (ERN; Gehring, Goss, Coles, Meyer, & Donchin, 1993) has been particularly important to this field of study. The ERN is a negative deflection maximal at fronto-central sites that occurs within 100 ms of an erroneous response. Source localization and functional magnetic resonance imaging (fMRI) studies have indicated that the ERN is generated by the anterior cingulate cortex (ACC; Carter, Braver, Barch, Botvinick, Noll, & Cohen, 1998; Gehring & Knight, 2000; van Veen & Carter, 2002). However, patient lesion studies have also implicated the lateral prefrontal cortex (PFC) in the generation of the ERN signal (Gehring & Knight, 2000; Stemmer, Segalowitz, Witzke, & Schönle, 2004; Ullsperger, von Cramon, & Muller, 2002). More recently, evidence from patients with frontal lobe white matter damage, but with intact ACC and lateral PFC gray matter, suggested that the combined functioning of these two

regions is important to the generation and propagation of the ERN (Hogan, Vargha-Khadem, Saunders, Kirkham, & Baldeweg, 2006). A smaller negative peak is found in correct response trials, termed the correct-response negativity (CRN), and is considered to represent an evaluative process (Vidal, Hasbroucq, Grapperon, & Bonnet, 2000), and be influenced by degree of stimulus and/or response uncertainty (Coles, Scheffers, & Holroyd, 2001; Falkenstein, Hoormann, Christ, & Hohnsbein, 2000; Pailing & Segalowitz, 2004). The difference between the amplitudes of the CRN and ERN provides an index of the functional integrity of the error-monitoring system (Falkenstein, 2004).

A third response-locked ERP component has been linked to performance monitoring. The error positivity (Pe) is maximal at centro-parietal sites and peaks at ~300 ms after an error response (Falkenstein et al., 2000; Nieuwenhuis, Ridderinkhof, Blom, Band, & Kok, 2001). The Pe reflects later error-processing activity that is independent of the processing linked to the ERN (Falkenstein et al., 2000, for a review of the functional significance of the Pe, see Overbeek, Nieuwenhuis, & Ridderinkhof, 2005). In support, studies have found that the ERN is generated by the caudal part of the ACC and the Pe by the rostral part (Herrmann, Rommner, Ehrlis, Heidrich, & Fallgatter, 2004; van Veen & Carter, 2002). However, the neural generators of the Pe are far from well understood, and other regions have been implicated in the Pe signal (e.g., O'Connell et al., 2007). Nonetheless, Herrmann et al. (2004) concluded, on the basis of different neural generators, that the ERN and Pe represented different aspects of error processing. The Pe has been linked to error recognition (Davies et al., 2004; Nieuwenhuis et al., 2001), however, this component is not well understood and there is continued debate over the functional significance of the Pe (see Overbeek et al., 2005).

The ERN and Pe, as correlates of error recognition, could potentially provide helpful neural indicators of cognitive development. However, Davies et al. (2004) using a visual flanker task with letter stimuli, found only small ERN components in a sub-sample of children aged between 7 and 12 years in their study, an increase in the amplitude of the CRN between the ages of 7 and 10 years. Pe amplitudes were not significantly different over the full age range of 7–18 years. On the basis that the amplitude of the ERN shows little age-related differentiation in childhood, later studies have included children aged 10 years and over (e.g., Santesso, Segalowitz, & Schmidt, 2005, 2006). Despite finding small ERN amplitudes in children, Davies et al. (2004) emphasized that there was substantial within-age group variability of the ERN amplitude, and reported robust ERN components in some of their children aged less than 10 years. Within the adult literature,

increased error rates have been linked with reduced ERN amplitudes (e.g., Hajcak, McDonald, & Simons, 2003; Herrmann et al., 2004). However, Wiersema, van der Meere and Roeyers (2007) failed to find a relationship between error rates and ERN amplitudes in their participants aged 7–8, 13–14, and 23–24 years. Thus, in younger children, the ERP correlates of error monitoring may not indicate a functional relationship between the neural basis of the ERN and cognitive performance. Instead, the ERN in younger children might reflect individual differences in the maturation of underlying neural substrates and/or the development of associated cognitive functions (Segalowitz & Dywan, 2008), either of which may also be reflected in the variability of RT.

The first aim in the current study was to investigate whether the ERN and post-error slowing differed in 7 and 9 year-olds by employing a large sample of children. Davies et al. (2004) included 124 participants aged 7–18 years, of which 12 children were aged 7 years and 18 children were aged 9 years. Based on the results of Davies et al. (2004), similar ERN and Pe amplitudes and post-error slowing between these age groups were expected. The second aim of the current study was to investigate factors which modulate error processing in children by concurrently considering both neural and behavioral indicators. The current study focused on 7 and 9 year-olds as there are significant behavioral and cognitive differences between these ages (Anderson, Reid & Nelson, 2001; Ridderinkhof et al., 1997). To date, there are no published papers investigating the relationship between intraindividual variability and the ERN in children though individual differences in response control have been investigated as correlates of the ERN amplitude in adults (e.g., Mathewson et al., 2005). It is plausible to hypothesize that whatever processes are influencing childrens' greater variability on behavioral measures, these will also be reflected in the neural error-monitoring signal.

METHODS

Approval for the study was provided by the Ethics Committee of the School of Psychology, University of Western Australia, Australia. Written informed consent was provided by each child and their parent or legal guardian.

Participants

Children aged 7–9 years, recruited from local schools, participated in Project K.I.D.S. (Kids Intellectual Development Study), a two-day holiday activity program investigating the cognitive, emotional, and social development of children. After excluding cases where performance was not significantly greater than chance for the congruent flanker stimuli (see below; 7 year-

olds: $n = 28$; 9 year-olds: $n = 14$), the final sample consisted of 36 children aged 7 years (17 male; $M = 7.54$ years, $SD = .27$) and 41 children aged 9 years (17 male; $M = 9.34$ years, $SD = .26$). There were no significant differences between those excluded and the current sample with regard to WISC-III full-scale IQ [$F(1, 117) = 2.61$, $p = .109$, partial $\eta^2 = .02$]. There were no significant differences between the 7 and 9 year-old samples on full-scale IQ as assessed using the WISC-III [$M = 115.2$, $SD = 11.7$ and $M = 113.6$, $SD = 12.8$ for the 7 and 9 year-old groups, respectively; $F(1, 75) = .32$, $p = .575$, partial $\eta^2 = .00$, $d = .13$]. However, there were significant age group differences on the Cattell Culture Free Intelligence Test raw scores [$M = 25.1$, $SD = 6.2$ and $M = 31.1$, $SD = 5.7$ for the 7 and 9 year-olds, respectively; $F(1, 72) = 28.49$, $p < .001$, partial $\eta^2 = .28$, $d = 1.24$], and the percentage of perseverative errors in the Wisconsin Card Sorting Test [$M = 22.8$, $SD = 15.4$ and $M = 15.8$, $SD = 6.7$ for the 7 and 9 year-olds, respectively; $F(1, 73) = 6.84$, $p = .011$, partial $\eta^2 = .09$, $d = .61$]. All participants were healthy at the time of testing, had no reported history of neurological or psychiatric conditions, and reported normal or corrected-to-normal vision and hearing.

Experimental Procedure

The modified visual flanker task used was based on the child-friendly paradigm reported by Rueda, Posner, Rothbart and Davis-Stober (2004). Each stimulus consisted of an array of five fish presented on a blue background. An arrow on the body of the fish indicated direction and the target was the central fish. Participants were instructed to press a response button situated on a keyboard (red felt patches on the 'Z' and '/' keys) corresponding to the direction of the central fish. There were three conditions: in the congruent condition (.5 probability), the five fish were green with the flanker fish pointing in the same direction; an incongruent condition (.25 probability), where all the fish were also green, however, the flankers pointed in the opposite direction to the target; and a reversed condition (.25 probability), in which the flanker fish were congruent, but all five fish were red, and required a response in the opposite direction to the central fish. Each fish subtended $.9^\circ$ horizontally and $.6^\circ$ vertically with $.2^\circ$ separating each fish and were randomly presented for 300 ms with a 2,000 ms inter-stimulus interval. Following Rueda et al. (2004), the task was presented as a game in which the participants had to feed the hungry central fish. Speed and accuracy were equally emphasized. A practice block of 16 trials was administered to ensure the participants understood the task requirements. A total of 480 trials were presented in three blocks separated by 5-min intervals.

Electrophysiological Acquisition

The EEG was continuously recorded using an Easy-CapTM. Eye movement activity was measured with bipolar leads placed above and below the left eye. The EEG was amplified with a NuAmps 40-channel amplifier, and digitized at a sampling rate of 250 Hz using a linked-mastoid reference with a ground lead located at AFz. Prior to recording, impedances were below 5 k Ω . The ERP processing was conducted offline using NeuroScan software. Offline, the EEG recording was digitally filtered with a

.05–30 Hz zero phase shift band-pass filter (12 dB down). The vertical ocular electrodes enabled offline blink reduction according to a standard algorithm (Semlitsch, Anderer, Schuster, & Presslich, 1986). All of the data are retained using this method, as eye movement artifacts are corrected in the EEG prior to any data analysis.

Data Analysis

Following the analysis parameters of Davies et al. (2004), EEG data were divided into epochs from –600 to 800 ms synchronized to the behavioral response, baseline corrected from –600 to –400 ms. Data from the midline sites were analyzed and epochs containing artifacts exceeding 100 μ V were automatically rejected. Epochs were averaged by response type (correct and error for each condition: congruent; incongruent; reversed). The peak amplitude measure was calculated as the amplitude of the most negative point in the latency window from response onset (time 0) to 100 ms post-response for the ERN (following erroneous responses) and CRN (following correct responses). In addition, a peak-to-trough amplitude measure was calculated by subtracting the amplitude of the most positive peak preceding the response (from –100 to 0 ms) from the amplitude of the ERN/CRN peak amplitude to account for the potential influence of the preceding positivity (Davies et al., 2004). The results from the peak amplitude measure and the peak-to-trough measure were similar, thus, the analyses of the peak amplitude measures only are reported. The ERN amplitude did not differ across conditions [$F(2, 150) = 1.09, p = .349$, partial $\eta^2 = .01$], and there was no interaction between condition and age group [$F(2, 150) = 2.08, p = .129$, partial $\eta^2 = .06$], therefore, error trials across the congruent, incongruent, and reversed conditions were combined to maximize the number of trials in the averaged waveforms (7 year-olds: $M = 77.28, 42.7$ SD, range = 20–176; 9 year-olds: $M = 75.95, 40.8$ SD, range = 10–197). The ERN is maximal at FCz (Davies et al., 2004) and amplitudes and latencies were identified at this location.

Behavioral responses made less than 100 ms or more than 2,000 ms after stimulus presentation were excluded from the

analysis. These data were assessed for normality and values 2 SD above and below the age group mean were also excluded from analyses. All behavioral data were analyzed with 3 (condition factor: congruent, incongruent, reversed) $\times$ 2 (age group factor: 7 year-old, 9 year-old) ANOVA models. The intraindividual coefficient of variation (ICV) was calculated by dividing the SD of the reaction time by the mean RT for each participant to adjust for the relationship between intraindividual variability and mean RT (e.g., Stuss, Murphy, Binns, & Alexander, 2003). Post-error slowing was calculated by subtracting the mean RT for congruent trials following an error from the mean RT for correct trials and a mixed design 2 (accuracy factor: correct, error) $\times$ 2 (age group factor: 7 year-old, 9 year-old) ANOVA was conducted. Post-error slowing was only calculated for post-error congruent trials, as it was rare that incongruent/reversed trials would follow incongruent/reversed trials. The ERP data were also analyzed with mixed design ANOVA models. The ERN and CRN were examined together in a 2 (component factor: ERN, CRN) $\times$ 2 (age group factor: 7 year-old, 9 year-old) ANOVA for amplitude and latency, separately. The amplitude and latency of the Pe were examined in a 4 (location factor: Fz, FCz, Cz, Pz) $\times$ 2 (age group factor: 7 year-old, 9 year-old) model. Pearson correlations were conducted to assess relationships between the amplitude of the ERP measures and behavioral indices. Speed-accuracy trade-off was explored with correlations between error rate and RT. When the assumption of sphericity was violated, the Greenhouse–Geisser adjustment to the degrees of freedom was applied.

RESULTS

Response-Locked ERP Components

CRN–ERN Components. As indicated in Figure 1 and Table 1, ERN amplitude was significantly greater than CRN amplitude and evidenced by a main effect of response accuracy [$F(1, 75) = 28.89, p < .001$, partial $\eta^2 = .28$]. There was no main effect or interaction with

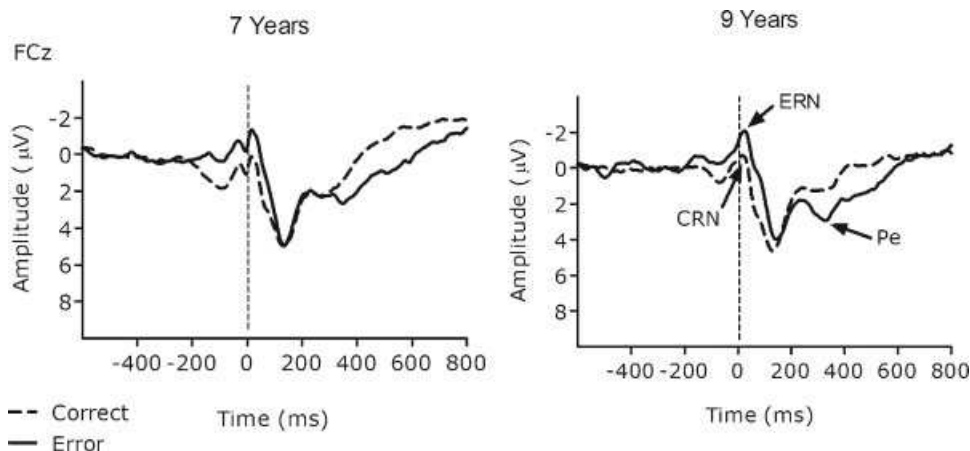


FIGURE 1 Grand group averages for the combined conditions correct and error waveforms in 7 and 9 year-olds at FCz.

Table 1. Peak Amplitudes and Latencies of Response-Locked Components

	7 Years (<i>n</i> = 36)		9 Years (<i>n</i> = 41)	
	Amplitude (μ V)	Latency (ms)	Amplitude (μ V)	Latency (ms)
CRN	-.5 (2.9)	20.1 (25.6)	-1.2 (4.2)	18.0 (22.6)
ERN	-2.1 (3.6)	19.0 (23.4)	-3.0 (3.4)	29.4 (26.9)
Pe	9.6 (5.5)	283.1 (63.9)	10.8 (6.8)	261.2 (60.3)

Note. Mean (SD).

age group, or gender with regard to the amplitudes of the CRN and ERN. These findings did not change when covarying for the number of trials within the correct and error waveforms [response accuracy: $F(1, 73) = 4.56$, $p = .036$, partial $\eta^2 = .06$]. However, the covariate of number of error trials was significant [$F(1, 73) = 6.95$, $p = .010$, partial $\eta^2 = .09$], and indicated that the number of error trials in the error waveform accounted for 9% of the variance in the amplitudes of the CRN and ERN. Post-hoc examination found that the covariate was no longer significant when the CRN and ERN were considered separately [$F(1, 74) = .65$, $p = .423$, partial $\eta^2 = .01$; $F(1, 74) = 1.08$, $p = .303$, partial $\eta^2 = .01$, respectively].

A main effect of response accuracy was also found for latency [$F(1, 75) = 4.40$, $p = .039$, partial $\eta^2 = .06$], revealing that the CRN occurred earlier than the ERN in both age groups.

Pe Component. A main effect of location [$F(3, 225) = 65.70$, $p < .001$, partial $\eta^2 = .47$] was found for Pe amplitude. Post-hoc examination found that Pz was the location of maximal amplitude, with significant differences between Pz and Fz, FCz and Cz [$F(1, 75) = 82.66$, $p < .001$, partial $\eta^2 = .52$; $F(1, 75) = 51.60$, $p < .001$, partial $\eta^2 = .41$; $F(1, 75) = 17.09$, $p < .001$, partial $\eta^2 = .19$; Fz, FCz, and Cz, respectively]. The amplitude and latency of the Pe did not differ significantly between the two age groups.

Behavioral Responses. As shown in Table 2, stimulus type influenced both error rate [$F(2, 150) = 90.35$, $p < .001$, partial $\eta^2 = .55$] and correct RT [$F(2, 150) = 65.18$, $p < .001$, partial $\eta^2 = .47$]. Post-hoc tests indicated that error rate was significantly increased in the reversed condition compared to the incongruent [$F(1, 75) = 10.95$, $p = .001$, partial $\eta^2 = .13$] and congruent conditions [$F(1, 75) = 204.98$, $p < .001$, partial $\eta^2 = .73$]. Correct RT was also significantly longer in the reversed condition compared to the incongruent and congruent conditions [$F(1, 75) = 20.49$, $p < .001$, partial $\eta^2 = .22$; $F(1, 75) = 143.45$, $p < .001$, partial $\eta^2 = .66$; incongruent and congruent, respectively]. Congruency effects on accuracy and correct RT were also found, in that accuracy was

reduced and correct RT was increased in the incongruent compared to the congruent condition [$F(1, 75) = 86.56$, $p < .001$, partial $\eta^2 = .54$; $F(1, 75) = 46.82$, $p < .001$, partial $\eta^2 = .38$; accuracy and correct RT, respectively].

In both groups, errors were faster than correct responses [$F(1, 74) = 183.05$, $p < .001$, partial $\eta^2 = .71$], regardless of condition. Main effects of condition and group were found on error RT [$F(2, 148) = 4.26$, $p = .016$, partial $\eta^2 = .05$; $F(1, 74) = 3.98$, $p = .050$, partial $\eta^2 = .05$; condition and group, respectively]. Post-hoc analyses revealed that error RT did not differ between the congruent and reversed conditions, but that there was a significant increase in error RT in the incongruent compared to the congruent condition [$F(1, 75) = 6.31$, $p = .014$, partial $\eta^2 = .08$]. Age group comparisons showed that error RTs were slower in the 7-year-old group than the 9-year-old group in the reversed condition [$t(66) = 2.27$, $p = .026$], and were numerically longer in the incongruent condition [$t(75) = 1.98$, $p = .051$].

A main effect of condition was found for correct response variability [$F(2, 150) = 4.41$, $p = .018$, partial $\eta^2 = .06$]. Post-hoc examination found that correct RTs were more variable for the incongruent and reversed conditions compared to the congruent condition [$F(1, 75) = 4.46$, $p = .038$, partial $\eta^2 = .06$; $F(1, 75) = 8.46$, $p = .005$, partial $\eta^2 = .10$; incongruent and reversed, respectively]. The standard deviation of reaction time (RTSD) to correct trials during the incongruent and reversed conditions did not differ significantly, nor was there a significant effect of age group on correct RTSD [$F(2, 150) = .48$, $p = .620$, partial $\eta^2 = .01$]. A main effect of condition for error RTSD was also found [$F(2, 150) = 9.17$, $p = .001$, partial $\eta^2 = .11$]. Post-hoc analyses indicated that error RT variability did not differ significantly between the congruent and incongruent conditions, and both were more variable than in the reversed condition [$F(1, 74) = 21.99$, $p < .001$, partial $\eta^2 = .23$; $F(1, 73) = 6.81$, $p = .011$, partial $\eta^2 = .09$; congruent and incongruent, respectively].

Speed-accuracy trade-offs were found in both groups for each condition (see Tab. 3). These correlations revealed that faster RTs were associated with reduced accuracy. In order to examine the effects of these speed-accuracy trade-offs in more detail, four groups were

Table 2. Error Rate, Error Reaction Time and Reaction Time Standard Deviation, Correct Reaction Time and Reaction Time Standard Deviation, and Post-Error Slowing in the Flanker Task

Condition	7 Years (<i>n</i> = 36)	9 Years (<i>n</i> = 41)	<i>d</i>
Error (%)			
Congruent	25.7 (10.5)	24.3 (11.1)	.12
Incongruent ^a	39.9 (16.3)	35.5 (15.7)	.28
Reversed ^b	42.4 (16.3)	42.5 (13.9)	.00
Correct RT (ms)			
Congruent	709.9 (199.1)	634.7 (176.2)	.41
Incongruent ^a	770.8 (233.3)	690.3 (207.5)	.37
Reversed ^b	801.1 (228.1)	751.6 (203.1)	.23
Correct RTSD (ms)			
Congruent	258.0 (85.3)	231.9 (90.4)	.30
Incongruent	280.1 (105.3)	239.1 (99.1)	.41
Reversed	283.8 (100.7)	257.0 (96.5)	.28
Error RT (ms)			
Congruent	640.0 (176.5)	582.0 (165.7)	.30
Incongruent ^a	680.0 (205.9)	590.4 (174.9)	.46
Reversed	661.9 (179.1)	577.3 (140.6) ^c	.53
Error RTSD (ms)			
Congruent	293.0 (119.5)	255.5 (112.5)	.33
Incongruent	287.9 (123.0)	229.7 (113.8) ^c	.50
Reversed	247.1 (105.6)	219.7 (100.5)	.27
Post-error slowing (ms)			
Congruent	16.6 (42.8)	10.8 (44.6)	.08

Note. Mean (SD).

^a*p* < .05 incongruent > congruent.

^b*p* < .05 reversed > incongruent.

^c*p* < .05 7 year-olds > 9 year-olds.

created based on median splits of correct RT and accuracy in the congruent condition. The groups (Slow_Low, Slow_High, Fast_Low, and Fast_High) did not differ with regard to full-scale IQ [$F(3, 73) = 1.50$, $p = .222$, partial $\eta^2 = .06$].

Relationships Between the ERN and Intraindividual Variability and Performance Measures

Of particular interest were the significant group-level correlations between ERN amplitude and error RT

Table 3. Speed-Accuracy Trade-Off Correlation Coefficients

	7 Years (<i>n</i> = 36)	9 Years (<i>n</i> = 41)
Congruent correct % and RT	.56	.48
Congruent error % and RT	-.59	-.50
Incongruent correct % and RT	.53	.65
Incongruent error % and RT	-.51	-.49
Incompatible correct % and RT	.74	.65
Incompatible error % and RT	-.72	-.71

Note. Pearson product-moment correlations; all $p < .01$.

variability averaged across the conditions [$r(77) = .34$, $p = .003$, see Fig. 2a]. The ICV was calculated to adjust for the relationship between intraindividual variability and mean RT. These findings were replicated with the ICV measure of intraindividual variability [$r(76) = .32$, $p = .004$, see Fig. 2b]. These correlations indicate that reduced ERN amplitude was associated with more variable error response times.

DISCUSSION

The aims of the study were to investigate error processing in children aged 7 and 9 years and the relationships between the ERN and intraindividual variability in reaction times. In support of Davies et al. (2004), the amplitude and latency of the ERN did not significantly differ between 7 and 9 year-olds. Consistent with this, amplitude and latency of the Pe also did not differ between the ages of 7 and 9 years. In sum, we have replicated the findings of Davies et al. (2004) using a larger sample.

The most significant finding in the current study was the relationship between reduced ERN amplitude and

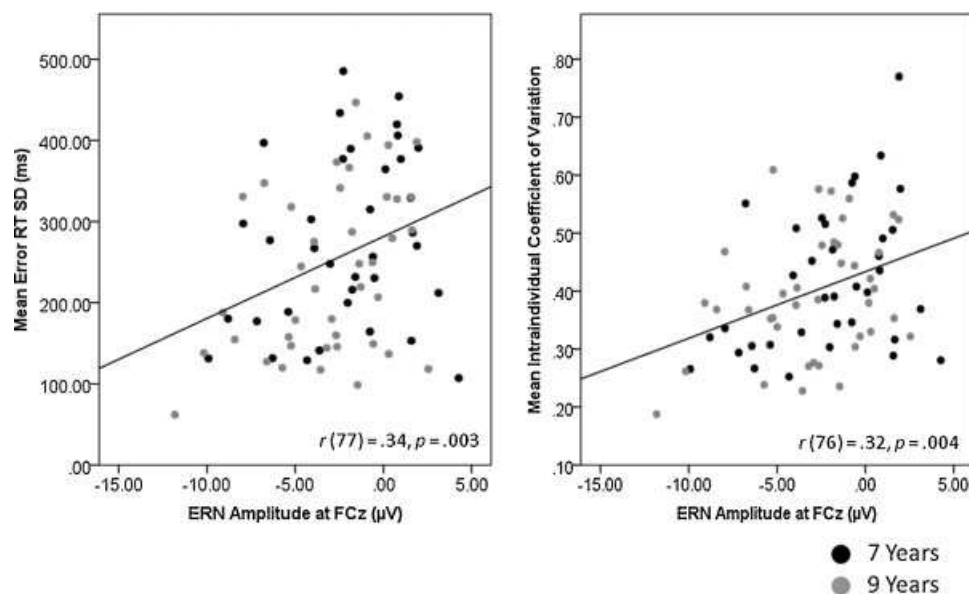


FIGURE 2 Scatterplots of the relationship between the ERN amplitude at FCz and (a) intraindividual variability (RTSD) and (b) the intraindividual coefficient of variation (ICV) across groups.

increased response variability in children aged 7 and 9 years. Greater error response variability represents wide fluctuations in responses, with responding including fast and slow errors. Fast errors may be due to impulsive responding, whereas slow errors may be caused by lapses in attention (Castellanos et al., 2005; Unsworth et al., 2010; Weissman et al., 2006), and/or failed inhibition of incorrect response tendencies (e.g., Bellgrove, Hester, & Garavan, 2004). Reduced inhibitory control was suggested by the findings that slower correct response times with increased condition complexity did not support equivalent accuracy in the conditions in the current study. Lapses of attention and reduced inhibitory control are not mutually exclusive explanations for intraindividual variability, as lapses of attention occur during stimulus processing, whilst inhibitory control is involved during response preparation and execution. A more complex design than the one used in the current study is required to separate these different processes. It would be of interest to compare performance on tasks that manipulate inhibitory control (e.g., Davidson et al., 2006) with the ERN elicited by different paradigms (e.g., Santesso & Segalowitz, 2008), and include groups presenting with high response variability and poor attention and inhibition (e.g., attention deficit hyperactivity disorder, ADHD; Castellanos & Tannock, 2002) to assess these relationships in more detail.

The finding of a relationship between the ERN and intraindividual variability has implications for the dominant theories of the functional significance of the ERN. The error-detection model states that the ERN represents a

monitoring process; an error signal is produced when a mismatch between the intended correct response and the actual response is detected (e.g., Coles et al., 2001; Falkenstein et al., 1991, 2000; Gehring et al., 1993). The conflict-detection theory contends that the ERN represents a measure of increased by stimulus–response conflict (Botvinick, Braver, Barch, Carter, & Cohen, 2001; Carter et al., 1998). The reversed condition in the current study provided an opportunity to assess response conflict. The ERN amplitude did not differ significantly between the conditions in the children, indicating that the reversed condition did not elicit greater amplitudes, supporting the view that the ERN is a generic error detection signal. This proposition is further supported by the finding that the amplitude of the ERN was significantly greater than that of the CRN; this difference represents a functioning internal error detection system (cf., Falkenstein, 2004). However, the finding of a relationship between error response variability and the ERN does not fit well with either of these theories. The reinforcement learning theory (Holroyd & Coles, 2002) posits that the ERN is generated by the ACC when it receives a negative learning signal sent by the dopamine system via the mesencephalic pathway (see also Jocham & Ullsperger, 2009). Reduced levels of dopamine have been linked with an increased signal-to-noise ratio in the brain, a neural correlate of intraindividual variability (MacDonald, Nyberg, & Bäckman, 2006). We speculate that increased signal-to-noise ratio in neural networks could contribute to reduced amplitude of the ERN in the children in the current study.

It should be noted that structural changes to the brain occurring during childhood could also influence the morphology of the ERN in children. The ERN has been localized to the ACC, and whilst similar activation of the ACC has been found in 5–16 year-olds (Casey et al., 1997), the lateral PFC and white matter have been implicated in the generation and propagation of the ERN (Gehring & Knight, 2000; Hogan et al., 2006; Ullsperger et al., 2002). Thus, the small ERN in children could reflect the protracted development of the lateral PFC and white matter in this age group. Evidence for this proposition is suggested by the findings that gray matter volumes increase through childhood and peak in the PFC at age ~12 (Giedd et al., 1999), and white matter volumes continue to increase from age 4 into middle age (Giedd et al., 1999; Paus et al., 2001; Sullivan, Rosenbloom, Serventi, & Pfefferbaum, 2004). Furthermore, recent studies have found more diffuse patterns of functional activation and connectivity in the PFC in children compared to adolescents and adults (Clare Kelly et al., 2009; Durston et al., 2006), which may influence the propagation of the ERN signal in children. The relationship between ERN amplitude and intraindividual variability in the current study might reflect the shared underlying neural substrates for these two indices of brain maturation.

Surprisingly, there were few behavioral differences between the 7 and 9 year-olds. Accuracy and correct RT in each of the conditions were not significantly different between the groups. There were no interactions with age group, indicating that the reduction in accuracy and the increase in correct RT with condition complexity also did not differ between 7 and 9 year-olds. Cragg, Fox, Nation, Reid and Anderson (2009), using a go/no go paradigm, also found that RTs did not differ significantly between 7- and 9-year-old children. The lack of correct RT differences between these groups of children is in contrast to a number of studies (e.g., Davidson et al., 2006; Davies et al., 2004; Kail, 1991; Ridderinkhof et al., 1997), which have typically found that RT is longer in younger children. Of note, the current study found that the mean RTs in the 7 year-olds were in the expected direction. The small effect sizes for RT and RTSD compared to those for fluid intelligence and perseverative errors suggested that the even with the good sample size our study might still be underpowered, and perhaps the lack of significant differences in RT and RTSD were in addition due to the relative insensitivity of these measures. The use of a child-friendly assessment paradigm may have also reduced any age-related differences in performance due to the active engagement of the 7 year-olds (Hogan, Vargha-Khadem, Kirkham, & Baldeweg, 2005).

One of the prominent caveats in the literature regarding factors that modulate the ERN is that increased error rate

is associated with reduced ERN amplitude (Hajcak et al., 2003; Herrmann et al., 2004). Although the error rates differed between the conditions, there were no significant differences in the amplitude of the ERN elicited by each condition (see also Davies et al., 2004). The conditions were combined in order to increase the number of error trials in the averaged waveforms for each groups, and the ERN in the combined condition waveforms was not related to error rates in any of the conditions, a finding shared with Wiersema et al. (2007). Although the number of error trials explained 9% of the variance in the amplitudes of the CRN and ERN, when these components were considered separately, the number of error trials did not influence the amplitude of either component. These results were supported by a non-significant correlation between ERN amplitude and the number of error trials in the error waveforms [$r(77) = .12, p = .296$]. These results are in contrast to Davies et al. (2004) who found that the number of error trials influenced the amplitude of the ERN. Thus, in our study, the ERN in children was not related to better performance on the task.

The current study replicated the finding that there were no developmental differences in the ERN and Pe in children aged 7 and 9 years using a larger sample. Significant amplitude differences were found between the CRN and ERN, indicating that both 7 and 9 year-olds processed errors differently to correct responses. This difference provides evidence for a functioning internal error detection system (cf., Falkenstein, 2004), and indicated, despite small ERN amplitudes in both groups, some degree of internal error processing in 7 and 9 year-olds. However, in contrast to previous studies (Davies et al., 2004; Wiersema et al., 2007), the current study provided little evidence of behavioral adjustment in response to errors, as indexed by post-error slowing.

In conclusion, the current study found that the amplitude of the ERN in 7 and 9 year-olds was related to intraindividual variability in the speed of responding. Increased intraindividual variability has been linked with lapses of attention, thus the results suggest that attentional lapses influence neural indices of error monitoring in children. In support of Davies et al. (2004), the results did not support developmental changes in error processing, as indexed by ERN and Pe amplitudes between 7 and 9 year-olds, and highlights the importance of considering intraindividual variability in studies exploring the developmental trajectory of error processing.

NOTES

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